MIL-T-13333A(Ord)
1 September 1959
SUPERSEDING
MIL-T-13333 (Ord)
23 March 1959

MILITARY SPECIFICATION

TELESCOPES, M48 AND M49

1. SCOPE

1.1 This specification covers two models of telescopes for observing effectiveness of artillery fire, These two telescopes are almost identical optically, but differ slightly in their construction.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-T-13123	Telescopes M48 and M49; Packaging of
MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly, and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

Class 22, Division 345	Observation Telescope, M48
Class 22, Division 346	Observation Telescope, M49

FSC 6650

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(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model. - At the beginning of production, one telescope, representative of production, shall be made available to the Government to determine the ability of the telescope to meet all requirements of this specification.

3.2 General specifications. - The telescope shall meet the following requirements of Specifications MIL-F-13926 and MIL-O-13830.

3.2.1 MIL-F-13926

(a) Order of precedence
(b) Protective finished surface
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.2.2 MIL-O-13830

(a) Cleanliness

3.3 Material. - Material shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.4 Design and construction. - Design and construction of the telescope shall be in accordance with the applicable Ordnance drawings as follows:

(a) Class 22, Division 345 for Telescope, Observation, M48
(b) Class 22, Division 346 for Telescope, Observation, M49

3.5 Environmental

3.5.1 Storage temperature. - The telescope shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all requirements of this specification at room temperature (60°F to 90°F) after having been exposed and thermally stabilized at ambient temperatures of minus 80°F and plus 160°F.

3.5.2 Operating temperatures. - The telescope focusing adjustment shall move freely while the telescope is exposed and thermally stabilized at ambient temperatures of minus 40°F and plus 150°F.

3.5.3 Vibration. - There shall be no visible damage or loosening of assembled parts as a result of vibration in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration, the telescope shall

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meet all of the following requirements of this specification.

3.6 Accuracy

3.6.1 Definition. - The resolving power at the geometric axis for a parallel bar test target shall be 3-1/2 seconds of arc or better.

3.6.2 Eyepiece. - Eyepiece shall focus on objects from 50 feet to an indefinite distance from the observer.

3.6.3 Image tilt. - The image of a plumb line shall be erect within 45 minutes of arc.

3.7 Identification marking. - Identification marking, unless otherwise specified, shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Contractor's quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

The contractor will not be restricted to the inspection station and inspection method of the written description provided that the substitutes thereof will provide, as a minimum, the quality assurance required by the contractual documents. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedures of applicable drawings and specifications shall be used. The contractor shall notify the Government of and

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obtain approval for any change to the written procedure that may affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Classification of tests. - The inspection and testing of the completed Telescopes M48 & M49 shall be classified as follows:

(a) Production model test
(b) Acceptance test

4.3 Production model

4.3.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

4.3.2 These tests shall be conducted at the contractor's facility using test equipment designed or procured for use on the remaining items.

4.3.3 The model shall meet all the requirements of this specification using the inspection methods prescribed herein.

4.3.3.1 Disposition of rejected Production model. - Should the production model fail to pass the test for Government acceptance, production of those components causing rejection will cease until the defective characteristics have been corrected in the production process. When corrections in the process have been made, a new production model shall be completely inspected by the Government. The rejected production model shall be corrected, where practicable, and shall become part of the first production lot.

4.4 Inspection provisions. - Quality assurance provisions common to all fire control instruments as specified in Specifications MIL-O-13830 and MIL-F-13926 shall apply where applicable.

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4.4.1 Inspection of components. - All positional and functional characteristics shown on the drawings contained in the contract shall be included in the inspection plan and shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements effecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a critical requirement. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein, that have a bearing on the functional accuracy and performance of the M48 and M49 Observation Telescopes.

4.4.1.1 Sampling inspection. - Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - over	30	1	2

4.4.1.1.1 Disposition of rejected lots. - Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection of the characteristics that caused the lot to be rejected.

4.4.1.2 Control of materials. - Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.4.2 Acceptance testing. - Sampling of the telescopes for acceptance may be used in lieu of 100 percent inspection and shall be in accordance with Standard MIL-STD-105. The characteristics listed below shall be inspected using the sampling plan specified below. Environmental test shall precede remaining acceptance tests and sample used shall be included in sample for remaining tests.

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(a) Definition
(b) Eyepiece focus
(c) Image Tilt
(d) Identification
(e) Cleanliness
(f) Workmanship

For normal inspection use AQL value 1.5 percent defective, inspection level II, sampling plan table IV-A of Standard MIL-STD-105.

Reduced inspection to be the same as normal inspection except inspection level I shall apply.

The environmental characteristics listed in Section III shall be inspected using the AQL value of 6.5 in conjunction with inspection level L-8 and sampling plan table IV-A of Standard MIL-STD-105.

4.4.3 Lot formation. - Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.3.1 Disposition of rejected lots. - Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

4.4.4 Inspection equipment

4.4.4.1 Contractor provisions. - Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.4.4.2 Accuracy requirements. - Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Specification MIL-P-13123.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selections of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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